NIHR | National Institute for Health and Care Research

Reference number	NIHR208140
Programme	Research for Patient Benefit
Call	RfPB URD & Specialisms highlight notice: Allied Health Professionals (Stage 2)
Lead Applicant	Dr GRAEME O'CONNOR
Host organisation	Great Ormond Street Hospital for Children NHS Foundation Trust
Research Title	Single-test accuracy study to assess the diagnostic value of faecal calprotectin to diagnose necrotising enterocolitis in infants with heart defects
Start Date	01/05/2025
Grant Duration (Months)	24
End Date	
Research Costs	£207,294.00
Research Type	Primary Research

Pre-submission

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1. The Research Team

Specify your (lead applicant) role in the research
Graeme O'Connor will take responsibility for ensuring that the project is successfully delivered. Graeme is a quantitative early career researcher who has led and published several small studies and is ready to embark on a larger-scale project. Graeme has experience and expertise in Patient and Public Involvement, he led the JLA Research Priority Partnership in nutrition and paediatric intensive care. Graeme will be supported by Professor Katherine Brown, an experienced paediatric cardiologist researcher. Graeme will manage the study at all four sites, including the day-to-day running (study set-up, ethical approval, data collection) and will also co-lead the PPI element.

% FTE Commitment: This refers to the percentage of your time that you will commit to this project. If you are funded as part of other NIHR projects that will be running concurrently, your time must not exceed 100% overall.
30

Joint Lead Applicant

Full Name	Professor Katherine Brown	**Position**	Professor Paediatric Cardiac Intensive Care
Department		**Institution**	NIHR Great Ormond Street Biomedical Research Centre

Relevant expertise and experience of Joint Lead Applicant
The joint lead applicant Professor Katherine Brown is an experienced Consultant in Paediatric Cardiac Intensive Care at Great Ormond Street Hospital and Deputy Head of Department of the Children's Institute of Cardiovascular Science at UCL. Professor Brown has experience leading national multicentre studies and has a far-reaching professional network and will was the lead investigator in the pilot study that preceded this application and was supported in that role by Professor Brown. National audit mortality cardiac Professor Brown will supervise, mentor and support Dr O'Connor who is an early careers researcher, to deliver all aspects of the research and will also contribute to the study steering group. Dr O'Connor was the lead investigator in the pilot study that preceded this application (faecal calprotectin in infants with cardiac defects) and was supported in that role by Professor Brown. Professor Brown's has research time in her existing contract. The requested FTE for this study will ensure commitment to support the delivery of this large multi-site study and will contribute to the network and dissemination of findings.

Specify role in research
Co-lead Prof Katherine Brown will take responsibility of the study ensuring key milestones are achieved including: support Dr O'Connor HRA and Ethical approval for all four sites; adhere with GOSH research governance; monitor recruitment target; patient safety and appropriate recording of adverse events; representation at international committees to build interest and awareness of study. Supervise, mentor and support Dr O'Connor who is an early careers researcher.

%FTE commitment	10

Co-applicant role and % FTE commitment

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Specify the role and percent FTE commitment of each co-applicant.

Full Name	Ms Deborah Ridout	**Institution**	UCL Great Ormond Street Institute of Child Health

Please indicate if this co-applicant is a member of the public
No

Specify role in research
Mrs Deborah Ridout will lead the statistical support: Sample size calculation Logistic regression analysis will be used to assess the univariate relationship between calprotectin concentration and NEC diagnosis, explored using receiver operating characteristic analysis and estimate area under curve to distinguish between NEC and No NEC. Linear regression analysis will be performed to assess relationship between faecal calprotectin concentration and other factors. Deborah will be a member of the Study Steering Group.

% FTE Commitment: This refers to the percentage of your time that you will commit to this project.
10

Full Name	Professor Nigel Hall	**Institution**	University of Southampton

Please indicate if this co-applicant is a member of the public
No

Specify role in research
Professor Hall, a consultant surgeon, research interests include necrotising enterocolitis. He will led the independent adjudication review for the diagnosis of NEC: 'blind endpoint review committee'. At the end of the patient's participation, a NEC diagnosis will be independently adjudicated by the 'blind endpoint review committee' consisting of a paediatric surgeon and cardiac intensivist both of whom will not be involved in patient care. Professor Hall will be a member of the Study Steering Committee

% FTE Commitment: This refers to the percentage of your time that you will commit to this project.

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Please indicate if this co-applicant is a member of the public

Yes

Please include a clear description of their role and the reasons why a public co-applicant is joining the team. Co-applicants who are patients, service users or carers are not obliged to complete a standard CV but are required to provide a summary of any knowledge, skills and experience relevant to their role in the application.

This is a national multicentre study that involves one of the most medically complex cohort of infants and therefore we feel essential to have a strong PPI lead that can bridge the lay and research world ensuring we consider all aspects of this important study

Experience: Mrs Gardner has lived experience as a parent of an infant born with major cardiac defects and spent many months on the cardiac intensive care ward at Great Ormond Street Hospital. Her son who is now 5yrs old still requires regular follow up and surgical intervention. Mrs Garner also has a second child who is 1 years and has Trisomy 21. Mrs Gardiner has previous PPI experience, providing support and guidance throughout our pilot study, involved in the design and outcome measures, lay summary and patient information sheets.

Mrs Gardner works full time in a senior human resources role in London, which has many transferable skills that will compliment her PPI Co-applicant role including organisational, interpersonal; communication; professional and kindness. Mrs Gardner is also confident and assertive, willing to challenge researchers/ clinicians when necessary. Has Disclosure and Barring Service clearance

Mrs Gardiner will be part of the Study Steering Group (SSG) and will attend all six meetings over the study period. Mrs Gardner will be provided with a GOSH Research Passport Honorary contract, covering indemnity insurance for the duration of the role; a security card and an institutional email address; access to project shared drives; staff development and training events related to the study topic:

SSG: This includes preparation and attendance for each meeting. Direct time and indirect time commitment. PPI Pre-SSG meeting will be chaired by PPI Co-Applicant.

Key support alongside SSG will be the development of patient documents - Lay summary/ patient information sheets/ End of study lay summary for participant families. Unforeseen study problems may need participating in a focus group to provide feedback on a proposal or reviewing a short guidance document.

Feedback: Assessing and reporting progress will be provided using the Elspeth Mathie Facilitating Feedback Model.

Support for Mrs Gardner outside of the research team will include: Deirdre Leyden PPI Manager GOSH and INVOLVE.

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Specify role in research

Mrs Gardner is an expert parent with lived experience of an infant with complex cardiac issues and has an excellent understanding of the practical real-life issues of an infant who required cardiac surgery and has extensive knowledge around enteral feeding and nursing. Mrs Gardner will be a member of the study steering group and will be our link between our PPI members at partner sites.

% FTE
Commitment: This refers to the percentage of your time that you will commit to this project.

0

2. History of application

Has this application been previously submitted to this or any other funding body?

No

3. Scientific abstract

Scientific abstract

Research question: Is faecal calprotectin an accurate diagnostic tool for necrotising enterocolitis (NEC) in infants with heart defects?

Background: NEC is a serious type of gut disease that can develop in infants born with cardiac defects, it causes inflammation and injury to the gut mucosa and may be life-threatening. Making an accurate and timely diagnosis of NEC is a significant clinical challenge. In the early stages of the disease the clinical presentation of NEC is difficult to identify correctly. An inaccurate diagnosis of NEC results in the withdrawal of an infant's milk feed and unnecessary administration of antibiotics, exacerbating parental stress in intensive care. Calprotectin is secreted into the lumen in the early stages of intestinal inflammation, thus elevated faecal calprotectin concentrations are evident in infectious and inflammatory diseases.

Aim: Assess the performance accuracy of faecal calprotectin to diagnose NEC in infants with cardiac defects

Objectives: Collect faecal samples to measure calprotectin levels before and after cardiac surgery, and at the same time the medical team are suspicious that NEC may be developing when using current 'Gold Standard' (clinical features, blood tests and gut X-rays).

Methods: A prospective, multicentre study (Alder Hey, Birmingham, Great Ormond Street and Southampton). Target sample size n=270 infants

Inclusion: Infants who require cardiac surgery or catheterisation for a major congenital heart defect.

Primary Outcome: Faecal calprotectin concentrations at a confirmed NEC diagnosis when using current 'Gold Standard'.

All infants recruited to the study will have stool collection (100mg) at baseline (hospital admission), and first stool post-surgery. The calprotectin will be measured using the commercially available enzyme-linked immunosorbent assay.

Index test: At the time the medical team suspect the presence of NEC the index diagnostic test (faecal calprotectin) will be collected to coincide with the reference test (gold standard). The calprotectin measurement collected at the time point when the medical team suspected NEC will be used to produce the Receiver Operating Characteristic (ROC) curves to assess the diagnostic ability and accuracy of faecal calprotectin to distinguish between NEC and no NEC.

At the end of the patient's participation, a NEC diagnosis will be independently adjudicated by the *'blind endpoint review committee*'. Retrospective secondary analysis incorporating the ROC calprotectin cut-off value to explore the impact of a false NEC diagnosis on the clinical and economic burden of diagnostic error for infants diagnosed with inconclusive suspected NEC.

Timelines: Total study period: 24 months; Recruitment period: 18 months

Anticipated Impact

Future validation study to assess the impact of adding calprotectin to current NEC diagnostic algorithm to measure a effectiveness to reduce diagnostic error - suspected NEC. Update international NEC clinical guidelines addition of calprotectin cut off levels to mitigate an incorrect NEC diagnosis and improve the timeliness and accuracy of making a NEC diagnosis.

Dissemination:

- End of study lay report to participating families

- Conference presentation and publication in journal

4. Plain English Summary

Plain English summary

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Our Aims:

1. Monitor levels of a special marker found in poo before and after heart surgery to identify if this marker is higher in infants who develop gut disease in babies born with heart defects.

2. Assess the impact of this special marker for diagnosing gut disease if added to the current tools available to doctors.

Background

Babies born with a heart defect often require lifesaving heart surgery. A serious type of gut disease can develop in babies born with heart defects because the heart is unable to pump enough blood to the gut, stopping it from working correctly.

It is difficult for doctors to diagnose gut disease in the early stages due to the limitations of the current way it is diagnosed, which includes clinical features, blood tests and gut X-rays.

This limitation in diagnosing gut disease means the medical team is unsure if gut disease is developing and incorrectly diagnose the baby as having gut disease. This results in the baby's milk feed being stopped for many days and given antibiotics unnecessarily. Repeatedly stopping the baby's milk feeds leads to poor growth and increases parents' stress.

We want to measure the amount of a special marker found in babies' poo to see if it relates to gut disease. Adding this special marker to the current available tests (clinical features, blood tests and gut X-rays) could improve the accuracy of diagnosing gut disease in babies born with serious heart defects allowing for better patient care.

In our previous small study, we found a relationship between this special marker found in poo with early stages of gut disease. However, to confirm our results, we need to measure this special marker in many more babies.

Design methods

1: Collect poo samples from babies admitted to the hospital before and after heart surgery. The collected poo samples will be delivered to the hospital laboratory to extract and measure the amount of this special marker.

2: To assess if this special marker is higher in babies with gut disease, we will collect a poo sample at the same time the medical team are suspicious that gut disease may be developing when using current tests available (clinical features, blood tests and gut X-rays).

If this special marker is linked to gut disease it will help doctors to give the correct treatment to babies.

Parent involvement in study delivery

Our Heart Advisory Parent Group will join the medical team as part of our steering group and will advise on all aspects of the study ensuring that the family and patient are at the forefront of any decision making.

Sharing the results of study

An end of study leaflet will be sent to participating families.

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5. Changes from first stage

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Dr Graeme O'Connor
By Email

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15 Church Street
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www.nihr.ac.uk

Wednesday 28 August 2024

Dear Dr O'Connor,

RfPB Under-represented disciplines and specialisms highlight notice: Allied Health Professionals: NIHR208140 - Single-test accuracy study to assess the diagnostic value of faecal calprotectin to diagnose necrotising enterocolitis in infants with heart defects

Following the Stage 1 review of applications submitted to the above competition, I am pleased to inform you that your application has been invited to progress to Stage 2 of the application process.

Further to our previous correspondence, I include your feedback from the Committee Assessors below. You are advised to address these feedback points in Stage 2 of the application process.

The Committee Assessors noted that this was a well written proposal with strong supporting rationale. However, there are several points on which the Committee Assessors have requested the research team's response:

- The list of questions, hypotheses, aims and objectives was considered to be unnecessarily complex and the applicants were asked to simplify the list.

- Concerns were raised regarding some aspects of the research plan and methodology, and the applicants were invited to address them in the Stage 2 application. Specifically, the applicants were asked to:
    - Clarify what sensitivity and specificity would be required for the test to be clinically useful.
    - Explain clearly the sample size calculation.
    - Describe the analysis plan for each objective.
    - Demonstrate how the recruitment rate would be feasible within the timeframe and include this in the Gantt chart.

- The applicants were asked to clarify what inclusive procedures would be adopted in the recruitment process to ensure that a representative sample would be achieved.

- The proposal would benefit from stronger patient and public involvement (PPI), potentially with the inclusion of a PPI co-applicant (e.g. a parent with relevant lived experience), and further details about how the PPI representative(s) would be involved throughout the study were requested.

- The Committee Assessors requested that the plain English summary (PES) be edited to avoid overgeneralization and replace the term 'dissemination' with a Plain English alternative. Please note that the word limit for a PES is 450 words.

- The Committee Assessors requested that the applicants consider whether there should be representation from other sites on the research team.

The Stage 2 application form is now available to complete on the Research Management System and the deadline for submission is **1pm on 09 October 2024.**

Please be aware that where applicable applications should be accompanied with the 'Funder Export' from the online SoECAT, obtainable via the NIHR Central Portfolio Management System (CPMS). Further guidance is available via the RMS application.

Please note that Open Access costs should not be included in the Detailed Budget costs of the Stage 2 application. These costs are already covered by an allocated Open Access Envelope, which is separate from the award value. The Open Access Envelope is specifically designated for covering the costs of making peer-reviewed research articles openly accessible, which directly result from the research funded by the award in question. For more details, please refer to the Open Access Funding Guidance.

Further information on the RfPB Programme, including guidance notes and frequently asked questions, can be found on the RfPB website. The RfPB Programme also encourages applicants to share the Committee Assessors feedback and speak with the NIHR Research Support Service (RSS) for further advice.

I wish you all the best for Stage 2 of the Competition.

Yours sincerely,

Ms Zuzanna Tajchman
Programme Manager

Changes from Stage 1

1. **The list of questions, hypotheses, aims and objectives was considered to be unnecessarily complex and the applicants were asked to simplify the list.**
   Aim and objectives simplified:

Aim 1: Assess the performance accuracy of faecal calprotectin in to diagnose NEC in infants with cardiac defects

Objective 1: Collect faecal samples to measure calprotectin levels before and after cardiac surgery, and at the same time the medical team are suspicious that NEC may be developing when using current tests available (clinical features, blood tests and gut X-rays).

Objective 2: Faecal calprotectin levels will be plotted using Receiver Operating Characteristic (ROC) curves to assess the ability of calprotectin to distinguish between NEC and no NEC.

Aim 2. Assess the clinical and economic burden of diagnostic error for infants diagnosed with inconclusive suspected NEC

Objective 3. Retrospective secondary analysis to explore the impact of a false NEC diagnosis in infants with suspected NEC

2. **Concerns were raised regarding some aspects of the research plan and methodology, and the applicants were invited to address them in the Stage 2 application. Specifically, the applicants were asked to:**

○ **Clarify what sensitivity and specificity would be required for the test to be clinically useful.**

The 'ideal biomarker' of neonatal infection and necrotising enterocolitis (NEC) should possess the following clinical characteristics:

- A well-defined optimal cut-off level that has a sensitivity and negative predictive value approaching 1.00, and specificity and positive predictive value >0.8. In contrast, a biomarker with high specificity and positive predictive value is good for 'ruling in' infection/NEC

Reference: Ng PC, Ma TP, Lam HS. The use of laboratory biomarkers for surveillance, diagnosis and prediction of clinical outcomes in neonatal sepsis and necrotising enterocolitis. Arch Dis Child Fetal Neonatal Ed. 2015 Sep;100(5):F448-52. doi: 10.1136/archdischild-2014-307656. Epub 2015 Jan 2. PMID: 25555389.

○ **Explain clearly the sample size calculation.**

To ensure faecal calprotectin is clinical useful we estimate an anticipated specificity of 0.8 with 5% precision, a threshold for the lower bound 95% CI of 0.75. This calculation assumes the expected rate of NEC cases classified according to the gold standard to be 7%, therefore we aim to recruit 270 infants to contribute to statistical analysis. We anticipate a True NEC rate of 7% (estimated n=19) and in addition the rate of inconclusive suspected NEC to be 15% (estimated n=40) and No NEC groups estimated n=211.

Given the observational nature of the study, lack of harm to the patient from participation and deferred consent, we expect few refusals and a low attrition rate (death), therefore assuming a loss to follow-up/drop-out rate of 10%, we plan to recruit 300 patients.

○ **Describe the analysis plan for each objective.**

Objective 1: We will describe demographic and clinical characteristics of our sample using mean, standard deviation or median, interquartile range for continuous data as appropriate and frequency, proportion for categorical data. We will check the representativeness of our sample by describing available characteristics for those patients who dropped out.

Exploratory analysis: Faecal calprotectin concentration pre and post heart surgery will be described and compared using paired t tests. Changes in concentration levels will be derived and described. Regression analysis will be used to explore the association between changes in concentration levels and the timing of the samples e.g. time of second sample relative to surgery time.

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Objective 2: Logistic regression analysis will be used to assess the univariate relationship between calprotectin concentration and NEC diagnosis. We will explore the data using receiver operating characteristic (ROC) analysis and estimate area under curve (AUC), sensitivity and specificity, along with 95% CIs, for cut-off points of calprotectin concentration to distinguish between NEC and No NEC. A ROC curve is a plot of sensitivity against (1−specificity), and shows the test's performance. The better the test, the closer the ROC curve approaches the top left-hand corner, which represents the combination of 100% sensitivity and 100% specificity.

We will describe and compare clinical characteristics and treatments over the post-op phase, such as number of days nil by mouth, antibiotic use and percentage feed volume target, for the NEC and No NEC groups of patients using chi square tests and t-tests as appropriate.

Linear and logistic regression analysis will be performed to assess relationship between faecal calprotectin concentration and other confounding factors such as weight for age Z-score, length of cardiac surgery time (cardiopulmonary bypass), pre surgery use of prostaglandin and antibiotics.

Objective 3: Healthcare and economic cost burden of diagnostic errors. Retrospectively assess if the addition of a calprotectin cut-off value could have improved patient outcomes in infants who were given an inconclusive diagnosis of suspected NEC. Outcome measures to assess patient burden on outcome measures, including the unnecessary use of antibiotic, number of days nil by mouth, length of stay and intravenous nutrition.

● **The proposal would benefit from stronger patient and public involvement (PPI), potentially with the inclusion of a PPI co-applicant (e.g. a parent with relevant lived experience), and further details about how the PPI representative(s) would be involved throughout the study were requested.**

This is a national multicentre study that involves one of the most medically complex cohort of infants and therefore we feel essential to have a strong PPI Co-Applicant who can bridge between the lay and research world to ensure we capture and consider all aspects of this important study.

Our PPI Policy

Sharing of power: **The PPI Team** will be part of the Study Steering Group (SSG) and will attend all six meetings over the study period. For SSG members people with lived experience, their individual healthcare experiences are their expertise. Respecting and valuing the knowledge of all those working together on the research is paramount. Support outside of the research team will include: Deirdre Leyden PPI Manager GOSH.

PPI Team:

PPI Co-Applicant and PPI Lead for GOSH: Mrs Hayley Gardner	PPI Cardiac Advisory Parent Group: PPI Lead from each centre (Alder Hay, Birmingham and Southampton
Co-Applicant Experience: Mrs Hayley Gardner has lived experience as a parent of an infant born with major cardiac defects and spent many months on the cardiac intensive care ward. Her son is now 5yrs old and still requires regular follow up at Great Ormond Street Hospital. Mrs Gardner also has a second child who is 1 years old and has Trisomy 21. Mrs Gardner has previous PPI experience, providing support and guidance throughout our pilot study in the design (outcome measures), lay summary and patient information sheets. Disclosure and Barring Service clearance. Mrs Gardner works full time in a Human Resources team in London and has many transferable skills that will compliment her PPI Co-applicant role including, organisational, interpersonal, communication, professional and	PPI Cardiac Advisory Parent Group Experience: Lived experience of having an infant in intensive care or an young person/ adult with a heart defect

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kindness. Mrs Gardner is also confident and assertive, willing to challenge researchers/ clinicians when necessary. Mrs Gardner will be provided with a GOSH Research Passport Honorary contract, covering indemnity insurance for the duration of the role; a security ID card and an institutional email address; access to project shared drives; staff development and training events related to the study topic.	
PPI Teams role within SSG meetings: This includes preparation and attendance for six meetings over 24 months. Direct SSG time: 3hr per meeting including 30mins PPI pre-SSG meeting to discuss agenda and action points of SSG to ensure key points are understood and clear. Responsibilities: review of research patient documents for example lay summary/ consent forms/ patient information sheets/ End of study lay summary for participant families. Unforeseen study problems may need participating in a focus group The PPI pre-SSG meeting will be chaired by PPI Co-Applicant (Hayley Gardner).	
PPI: Payment Policy PPI Co-Applicant (Mrs Gardner) will be registered as an Honorary Supplier and can therefore invoice the CI for an electronic payment. Budget has been completed with Mrs Gardner: *6 x £150 (£900) This includes preparation and attendance for one day meetings. Direct time: 3hr per meeting. Indirect time: 2-3hrs per meeting additional reading* *Private Childcare – (£50 x 6 meetings = £300) £5 per meeting to cover the cost of telephone calls, paper, printing ink and paper, internet connection and other home sundries (£45)*	*PPI: Payment Policy* PPI Cardiac Advisory Parent Group. Voucher payments for time: £60 per meeting (direct and indirect time). Reading preparation and comments of research documents. Total spend = £1080
PPI Team: Training and Development Training course: Homepage - Learning for Involvement NIHR Guidance on co-producing a research project (learningforinvolvement.org.uk) NIHR Co-Production in Action (One, two, three and four) - Learning for Involvement	
Feedback: Assessing and reporting progress will be provided using the Elspeth Mathie Facilitating Feedback Model. The role of patient and public involvement leads in facilitating feedback: “invisible work” | Research Involvement and Engagement | Full Text (biomedcentral.com)	

○ **Demonstrate how the recruitment rate would be feasible within the timeframe and include this in the Gantt chart.**

Response: Feasibility of recruitment based on National figures for 12 cardiac centres in the UK admission rates in 2022/23 NCHDA-Domain-Report_2021_FINAL.pdf (hqip.org.uk):

- *Paediatric Surgery undertaken using cardiopulmonary bypass =3,080* PLUS *Paediatric Interventional catheterisation procedures = 2,397= National Total (12 cardiac centres- UK and Ireland) = 5,477*
- *Therefore, expected recruitment potential for 4 centres (GOSH, Alder Hey, Birmingham and Southampton)*

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who are under 3 months old (our inclusion criteria) = estimated 600 infants

- *Four recruiting centres aiming for 18 infants per month over 18 months = 324 infants.* ***Target sample size 270 infants****; assuming a 10% attrition.*
- Recruiting target mitigating action: A deferred (waived) consent process improves trial recruitment numbers in research. Our study has adopted a deferred consent model to ensure stool samples are not missed upon infants arrival to the intensive care units. *Reference Armstrong, S., Langlois, A., Laparidou, D. et al. Assessment of consent models as an ethical consideration in the conduct of prehospital ambulance randomised controlled clinical trials: a systematic review. BMC Med Res Methodol* ***17****, 142 (2017). https://doi.org/10.1186/s12874-017-0423-4*
- *Recruitment targets have been added throughout Gantt chart.*

		Year 1 (2025/26)												**Year 2 (2026/27)**							
Project Management		M	J	J	Au	S	O	N	D	J	F	M	Ap	M	J	J	Au	S	O	N	D
Recruitment Target					18				108				180				216			300	

● The applicants were asked to clarify what inclusive procedures would be adopted in the recruitment process to ensure that a representative sample would be achieved.

Response: A national study led by our team at Great Ormond Street Hospital monitored minority groups in incidence of serious cardiac defects requiring cardiac intervention before 1 year of age. The study identified 5350 infants (2940 (55.0%) boys). Overall CHD incidence was significantly higher in Asian and Black ethnic groups compared with the White reference population (incidence rate ratios (IRR) (95% CIs): Asian 1.5 (1.4 to 1.7); Black 1.4 (1.3 to 1.6)); The birth prevalence of specific defects varies by racial or ethnic group, and British Asian children have been reported to be at higher risk of complex CHDs than non-Asians. Some authors have suggested that socioeconomic disadvantage and reduced access to diagnostic services underlie ethnic differences in CHD prevalence, and deprivation has been associated with higher risk of all congenital anomalies, including CHDs, in the UK.

Hoskote A, Ridout D, Banks V, Wray J, Brown K. Neurodevelopmental status and follow-up in preschool children with heart disease in London, UK. Arch Dis Child. 2021 Mar;106(3):263-271. doi: 10.1136/archdischild-2019-317824. Epub 2020 Sep 9. PMID: 32907808.

Mitigating actions to ensure the study recruits families from ethnic minority group:

- One of the main barriers to research participation is language barrier. Participants who may have difficulties in an adequate understanding of English will be supported by The NHS interpretation service. 'Thebigword' is the sole supplier of telephone interpreting, a service which will enable researchers to help any client who may have limited English language skills. 0800 694 509
- We have ringfenced £600 (35hrs) to cover costs for interpreting service
- Reassuringly, our published pilot data managed to recruit families from ethnic minority groups to reflect to CHD population. Therefore we feel confident this will continue in our proposed study
- Of note, our Equality, Diversity and Inclusion (EDI) Strategy sets out Great Ormond Street Hospital Children's commitment, vision and ambition to do more to create an inclusive research environment and workplace. 21_01_Charity_EDI_Strategy_Report_ST7.2.pdf (gosh.org). The research team is also familiar with NHS England's practice guide to increase diversity in research: Increasing diversity in research participation: A good practice guide for engaging with underrepresented groups (england.nhs.uk)

Mitigating Actions:

- Deferred consent – can aid inclusivity when used in practice but also to incorporate ethnic minorities where they may have been otherwise under-represented.
  Reference: Raven-Gregg, T., Shepherd, V. Exploring the inclusion of under-served groups in trials methodology research: an example from ethnic minority populations' views on deferred consent. Trials 22, 589 (2021). https://doi.org/10.1186/s13063-021-05568-z
- Additionally, a layered consent process, that gradually provides information about the study to build trust has also been found to be an advantage in recruiting families from an ethic minority group.
- Burden and inconvenience to attend research appointments are known to be another barrier to recruiting families from an ethic minority group and poor social economic background. A mitigating factor for this barrier will be that for our proposed study infants will be on the intensive care unit for the duration of study period so we can approach families at their convenience with no need for any additional research visits.
- Inclusivity of social-economic group: Patient information sheets reading level will be set at 10-12yrs
  Reference: Tamariz L, Palacio A, Robert M, Marcus EN. Improving the informed consent process for research subjects with low literacy: a systematic review. J Gen Intern Med. 2013 Jan;28(1):121-6. doi:

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10.1007/s11606-012-2133-2. Epub 2012 Jul 11. PMID: 22782275; PMCID: PMC3539038.

● The Committee Assessors requested that the plain English summary (PES) be edited to avoid overgeneralization and replace the term 'dissemination' with a Plain English alternative. Please note that the word limit for a PES is 450 words.

- Aims have been simplified and summary has been reworded to be less overgeneralised.
- Patient and Public involvement replaced with parent involvement in study delivery.
- Dissemination replaced with sharing results of study.

● The Committee Assessors requested that the applicants consider whether there should be representation from other sites on the *research team.*

Named collaborator at each site. This section has been extended in the Research Plan - expertise section

Alder Hey:

Dr Bernadette Khodaghalian (consultant cardiologist at Alder Hey): Bernadette.Khodaghalian@alderhey.nhs.uk

Head of Cardiology: Dr Caroline Jones: Caroline.Jones2@alderhey.nhs.uk

Research Nurse Lead: Mrs Laura Rad: Laura.Rad@alderhey.nhs.uk

Birmingham Childrens Hospital:

Intensive care research Lead: Mrs Helen Winmill: Helen <h.winmill@nhs.net>

Research nurse: Mrs Samantha Owen: samantha.owen10@nhs.net

Southampton Hospital:

Southampton: Consultant surgeon Prof Nigel Hall: n.j.hall@soton.ac.uk

Research lead nurse. Mrs Jenny Pond: j.pond@soton.ac.uk

6. Detailed Research plan

Using all of the headings in the order presented below, please use this section to clearly explain your proposed research. Schematics, tables, illustrations, graphs, and other types of graphics can be embedded to clarify the research plan but they should not clutter the central narrative. Images do not count towards the overall word count but inclusion of them to overcome word limits is not permitted. Images may only be included within the 'Research Plan.' Images included in other sections will be removed from the application and not seen by reviewers.

For further information for what is required under each heading please refer to the Guidance for Applicants. **As this is the main part of your application which will be considered by the reviewing committee, you should ensure that the information is accurate, succinct, clearly laid out and provides sufficient methodological detail.**

NOTE: Applicants should aim to reserve a significant proportion of the word count for the project plan to enable methodological approaches to be fully specified.

1. Background and Rationale
2. Aims and Objectives
3. Research Plan / Methods
4. Dissemination, Outputs and anticipated Impact
5. Project / research timetable
6. Project Management
7. Ethics / Regulatory Approvals
8. Project / Research expertise
9. Success criteria and barriers to proposed work

1. Background

Necrotising enterocolitis (NEC) is a serious gut disease that can develop in infants born with cardiac defects, it causes inflammation and injury to the gut mucosa and may be life-threatening.[1] The aetiology of NEC in cardiac infants is multifactorial, associated with poor cardiac outflow and poorly oxygenated systemic circulation resulting in suboptimal blood flow to the intestines.[2]

Pre-submission

Making an accurate and timely diagnosis of NEC is a significant clinical challenge.[3] The clinical presentation of NEC is difficult to identify correctly in the early stages of the disease. Of concern, infants who are diagnosed with NEC at a later stage have worse disease.[4] Due to the potentially life-threatening effects of NEC, clinicians who suspect a baby has NEC but are unable to confirm, temporarily manage infants as if they have NEC as a precautionary measure.

This leads to infants without NEC having their milk feeds stopped for several days and being given unnecessary antibiotics. Compared to infants with confirmed NEC, the care of infants with suspected NEC is widely variable in terms of antibiotic regimen chosen, length of antibiotic treatment, and length of time nil-by-mouth - all these variabilities may adversely contribute to the length of hospital stay.[5]

We want to study a method to improve the accuracy of diagnosing NEC. We want to measure a protein marker found in baby's stools that if combined with currently available tests (clinical features, lab tests and abdominal X-rays)[6,7] may improve the timeliness and accuracy of making a NEC diagnosis, crucially at an early stage of gut disease.

The protein marker is called calprotectin and monitoring levels in baby's stools before and after heart surgery may help us to understand its relationship to NEC. Calprotectin is a biomarker that is released by white blood cells and is present in the gut when it is inflamed and can be measured in stool.[8] The amount of calprotectin in the stool may indicate the level of gut inflammation and the presence of NEC in infants with heart defects.[9]

The addition of calprotectin measurement to the current routine parameters (clinical features, lab tests and abdominal X-rays) may improve the timeliness and accuracy of diagnosing NEC in infants with cardiac defects, hence improving their care pathway and outcomes.

Why is this research important in terms of improving the health of patients?

1. An inaccurate diagnosis of NEC results in the withdrawal of an infant's milk feed, exacerbating parental stress in intensive care. Improved nutritional support has been associated with increased survival and a reduction in overall morbidity in infants with cardiac defects.[10]

2. An inaccurate diagnosis of NEC results in the unnecessary administration of antibiotics. Antibiotics cause antibiotic-related diarrhoea, which in turn affects expected growth.[11,12]

3. An inaccurate diagnosis of NEC results in the inappropriate use of intravenous nutrition as a substitute for enteral feeds during gut rest. Avoiding intravenous nutrition reduces the risk of line sepsis and liver disease.[13]

4. An accurate and timely diagnosis of NEC will reduce the length of hospital stay. Infants classified with suspected NEC have a longer hospital stay.[5,9]

5. The current *per-patient* marginal cost due to NEC is £13,900 and the cost due to feeding difficulties after surgery is £5,500,[14] both of which could be reduced with a timely and accurate diagnosis of NEC, improving patient care.

Review of existing evidence

Importance to NHS and patients

Congenital heart disease (CHD) is diagnosed in approximately 1 in 100 births.[15] Significant CHD that required intervention in early childhood affected 3.6 per 1000 live births in England 2004-2016.[16] The unadjusted crude 30-day mortality rate amongst 9,586 paediatric cardiac operations undertaken between 2019-2022 was 2.1% but was highest in the youngest babies.[7]

The number one priority for the 'Research priorities in children and adults with congenital heart disease: a James Lind Alliance Priority Setting Partnership': *how can damage to the gut during heart surgery in children with heart defects be minimised to reduce complications?*[17]

Another priority listed in the same Priority Setting Partnership is '*What are the best approaches to feeding in babies with heart defects, to optimise growth, improve outcomes, and reduce parental distress?*'[17]

In our national prospective study of NEC in 1819 infants who had cardiac surgery in 5 of the 10 UK specialist paediatric cardiac centres we found 4% had NEC, with this proportion rising to 7.5% amongst neonates under 28 days of age.[18]

Pre-submission

Necrotising Enterocolitis

Infants with CHD are at high risk of developing NEC due to persistent diastolic flow reversal in the abdominal aortic, which reduces mesenteric blood flow.[19] The aetiology of NEC is multifactorial with the interaction of cardiac surgery, gut hypoxia, poor mucosal integrity, and disruption of intestinal microbial dysbiosis.[20,21]

Management of NEC involves supportive care, empirical antibiotic therapy, parenteral nutrition, and bowel rest with gastric decompression initiated as soon as NEC is suspected.[22] The exact duration of bowel rest in infants with non-surgical NEC is based on disease severity and clinical judgment.[23]

Diagnostic Limitations of Necrotising Enterocolitis

Interpretation of the clinical picture, including relevant plain abdominal radiographs in infants is challenging and is one of the reasons for a wide variation in the reported incidence rate of NEC, ranging from 3 to 18%.[5] Key markers of NEC including intestinal dilation, ileus and pneumatosis intestinalis are often absent on abdominal radiographs.[25]

The clinical presentation of NEC includes lethargy, temperature instability, recurrent apnoea, bradycardia, hypoglycaemia, and shock. Specific gastrointestinal symptoms include abdominal distention (70% to 98%), blood per rectum (79% to 86%), high gastric residuals after feeding (>70%), vomiting (>70%), and diarrhoea (4% to 26%).[24]

In an audit of 195 neonates at one cardiac centre (Great Ormond Street Hospital) 14% had feeds suspended during an evaluation for NEC for at least 24 hours but did not have a NEC diagnosis. This clinical path can involve inappropriate use of intravenous nutrition, unnecessary antibiotics administration and longer hospital stays.[5,9] Of note, optimal nutritional support in infants requiring cardiac surgery has been associated with increased survival and a reduction in overall morbidity.[10]

Faecal Calprotectin- Biomarker

Neutrophils are first-line responders of the innate immune system and generally combat infections by phagocytosis.[26] Calprotectin is a neutrophil activation marker and is mainly exhibited in the cytoplasm of neutrophils and expressed on activated monocytes and macrophages.[27] Calprotectin is secreted into the lumen in the early stages of intestinal inflammation, thus elevated faecal calprotectin concentrations are evident in infectious and inflammatory diseases such as inflammatory bowel disease.[28] Faecal calprotectin is routinely used as a screening test in the evaluation of inflammatory gastrointestinal diseases and a faecal calprotectin value above 50µg/g should prompt review and raised faecal calprotectin of greater than 200µg/g should prompt urgent assessment.[29]

Faecal calprotectin is used as a diagnostic marker for NEC in preterm infants.[30] However, infants with CHD pose a unique clinical presentation in that their gut may be inflamed from birth due to compromised gut perfusion and therefore warrant specific investigation.[5] Additionally, ischemia-reperfusion injury from cardiac surgery results in pronounced neutrophil activation followed by a hyperinflammation reaction that drives inflammation in NEC.[31]

Pilot Study

Our published prospective single-centre observational pilot study monitored the relationship between faecal calprotectin concentration and gut inflammation in 30 infants admitted with cardiac defects requiring cardiac surgery. Faecal calprotectin for infants who developed NEC (n=5) had a mean concentration of 3528µg/g (1004SD) compared with 1339µg/g (1008SD) in patients with suspected NEC (n=7) and 390µg/g (410SD) in those without NEC (n=18) (P=0.0001). Infants with suspected NEC had a significantly longer length of hospital stay, on average 18 days longer compared to infants without NEC (P=0.03).[9]

Outcomes

Research is warranted to establish whether faecal calprotectin can be a useful addition to the diagnostic algorithm for NEC in infants with heart defects.[5,9] Important areas for further work include NEC prognostication, which includes earlier identification of the need for care escalation and the requirement for surgery.[32] Considering the proposed different disease processes for cardiac infants, it is essential to investigate calprotectin levels independently from other diseases.[5,9,31]

2) Research aims and questions

Question: Is faecal calprotectin an accurate diagnostic tool for NEC in infants with heart defects?

Hypothesis: Faecal calprotectin is higher in infants who have NEC than infants who do not have NEC.

Pre-submission

Aim 1: Assess the performance accuracy of faecal calprotectin to diagnose NEC in infants with cardiac defects

Objective 1: Collect faecal samples to measure calprotectin levels before and after cardiac surgery, and at the same time the medical team are suspicious that NEC may be developing when using current tests available (clinical features, blood tests and gut X-rays).

Objective 2: Faecal calprotectin levels will be plotted using Receiver Operating Characteristic (ROC) curves to assess the ability of calprotectin to distinguish between NEC and no NEC.

Aim 2. Assess the clinical and economic burden of diagnostic error for infants diagnosed with inconclusive suspected NEC

Objective 3. Retrospective secondary analysis incorporating the ROC calprotectin cut-off value to explore the impact of a false NEC diagnosis in infants with suspected NEC on patient treatment burdens.

3) Research Plan/ Methods

Study design

A prospective, multicentre, cohort-based study.

Setting

Specialist children's hospitals undertaking cardiac surgery in England: Great Ormond Street Children's Hospital, Alder Hey, Birmingham and Southampton hospitals.

Inclusion Criteria

Infants up to 3 months old, who require admission to a specialist cardiac centre for treatment of major congenital heart disease. Require cardiac surgery or interventional catheterisation, including babies who are treated with invasive medical therapies of respiratory support or intravenous vasoactive medications and their pathway will include a cardiac intervention of any type.

Exclusion criteria

Patients will be excluded if they are older than 3 months, do not have major heart disease and do have any gastrointestinal abnormality including gastroschisis or imperforate anus. Also, patients with neutropenia will be excluded since calprotectin is not activated when there is neutropenia.

Data Collection

All participants will have calprotectin levels measured pre- and post-cardiac surgery. Calprotectin will also be collected and measured if the medical team believe NEC is developing/ present using current standard assessment (clinical assessments, laboratory tests and abdominal radiographic findings). Patients will remain in the study from admission until discharge to home or 30 days post cardiac surgery whichever is first.

Other data collection: diagnosis (cardiac and non-cardiac conditions), interventions received by the infant (surgery type and invasive medical treatments), cardio-pulmonary bypass time (min); weights (kg), at birth, admission, and discharge, Z-scores for weight-for-age and length-for-age.

Primary Outcome

- Faecal calprotectin concentrations at a confirmed NEC diagnosis using current tests available (clinical features, blood tests and gut X-rays).

Secondary Outcomes

- Number of days with no enteral feeds – nil by mouth (ascertained at 0600 hours).
- Number of days on 100ml/kg/hour or more of enteral feeds (ascertained at 0600 hours).
- The number of days to re-establish breastfeeding post-cardiac surgery.
- Patient length of stay in intensive care and hospital.
- Number of days on antibiotics.
- Number of days on parenteral (intravenous) nutrition.
- C-Reactive Protein (mg/l) and complete blood count.
- Formal radiological abdominal X-ray results.

Pre-submission

- Structured clinical evaluation of the abdomen.

Feeding Protocol: patients in the study will be fed based on local unit clinical guidelines for feeding.[33]

REDCAP will be used to collect data at each site. Data will be anonymised to allow for data sharing.

Consent

Given that no harm is expected with participation in this observational study, a deferred consent procedure will be used following processes used in prior PICU research.[34] Patients will be enrolled on arrival at the specialist centre soon after birth, at a time when parents have a huge amount going on. Thereafter, parents will be approached at a suitable time when their child is stable with written information about the study. Written informed consent will be obtained before discharge and if any parent declines consent their wishes will be respected by removal of their child's data from the study.

Aim 1. Monitor faecal calprotectin levels before and after heart surgery and assess the performance accuracy of faecal calprotectin in infants who develop NEC

Objective 1: All infants recruited to the study will have stool collection (100mg) at baseline (hospital admission), and first stool post-surgery. Stool samples will be collected and stored in plastic test tube containers and directly sent to hospital clinical laboratory. The calprotectin will be measured using the commercially available enzyme-linked immunosorbent assay (ELISA) kit of Bühlmann Laboratories AG (Switzerland) according to the manufacturer's instructions. In brief, monoclonal capture antibodies highly specific to the calprotectin heterodimeric and polymeric complex are coated on the microliter plate when calprotectin was assayed in a single measurement using DS2 automated ELISA processing system (extended range 30–1,800 µg/g).

A limitation identified from our pilot study is the unpredictability of the infant's bowels opening, therefore for each time point we will get the first available stool and in some infants, there will be delays in stool arriving.

Faecal calprotectin performance analysis to distinguish between infants who do and do not have NEC: Infants with suspected NEC will conform with the reference test (gold standard - clinical assessments, laboratory tests and abdominal radiography, the definition was developed for use in a national audit by a multi-disciplinary group).[6,7]

Reference test NEC diagnosis is the identification of a clinical syndrome involving features from 1) plus either 2) or 3) or both 2) and 3) as outlined below:

1) Systemic signs - Include temperature instability, apnoea, bradycardia, raised inflammatory markers, thrombocytopenia, and shock features. In NEC these are present with abdominal and or radiological signs stated below.

2) Abdominal - include abdominal distension, reduced or absent bowel sounds, larger than normal gastric aspirates, gastric bleeding, rectal bleeding, abdominal tenderness, or cellulitis.

3) Radiological signs - The *abdominal radiograph* will be performed in the supine anteroposterior position. The following radiography findings are considered typical for NEC:

- Dilated bowel loops (often asymmetrical in distribution)
- Pneumatosis intestinalis
- Portal venous gas
- Pneumoperitoneum

Grading classification for NEC:

- Mild based on features from the systemic and abdominal signs the child recovers after 5-7 days of conservative treatment.
- Severe a more serious picture with perforation and peritonitis.

Index test: Faecal calprotectin concentration

At the time the medical team suspect the presence of NEC the index diagnostic test (faecal calprotectin) will be collected to coincide with the reference test (gold standard- clinical assessments, laboratory tests and abdominal radiography).

Objective 2:

Establishing calprotectin cut-off values to diagnose NEC: The calprotectin measurement collected at the time point when the medical team suspected NEC will be used to produce the Receiver Operating Characteristic (ROC) curves to assess the diagnostic ability and accuracy of faecal calprotectin to distinguish between NEC and no NEC.

Pre-submission

The ROC analysis will provide two outcomes: the diagnostic accuracy of faecal calprotectin and the optimal cut-point value for faecal calprotectin. The identification of the cut-point value will require a simultaneous assessment of sensitivity and specificity. A cut-point will be referred to as optimal when the point classifies most of the individuals correctly – area under curve (AUC).

Adjudication of a NEC diagnosis - 'blind endpoint review committee'

At the end of the patient's participation, a NEC diagnosis will be independently adjudicated by the *'blind endpoint review committee*' consisting of a paediatric surgeon and cardiac intensivist both of whom will *not* be involved in patient care, a diagnosis will refer to a consensus-based definition (outlined above) using current gold standard-clinical assessments, laboratory tests and abdominal radiography.[6,7] They will be blinded to faecal calprotectin values. The *blind endpoint review committee* will independently adjudicate a diagnosis as either: No NEC or NEC. If there is a disagreement about the final diagnosis, cases will be reviewed and adjudicated in conjunction with a third physician with relevant expertise. Prof Nigel Hall will be one of the adjudicators, assigned as a co-applicant. The second adjudicator will be listed as a collaborator.

Aim 2: Assess the clinical and economic burden of diagnostic error

Objective 3. secondary analysis to retrospectively review infants where the medical team were unable to confidently confirm or reject NEC due to the limitations of the gold standard and therefore classified infants as having suspected NEC. Outcome measures to assess patient burden on outcome measures, including the unnecessary use of antibiotic, number of days nil by mouth, length of stay and intravenous nutrition.

Sample size:

For our study we aim to recruit 300 patients, to be able to estimate an anticipated specificity of 0.80 with 5% precision, i.e. a threshold for the lower bound 95% CI of 0.75, which we have defined as clinically acceptable. This calculation assumes the expected rate of NEC cases classified according to the gold standard to be 7%. Furthermore, given the observational nature of the study and lack of harm to the patient from participation, we expect few refusals and a low attrition rate (death), therefore assuming a loss to follow-up/drop-out rate of 10%, we expect a minimum of 270 patients to contribute to our analysis.

Statistical Analysis:

We will describe demographic and clinical characteristics of our sample using mean, standard deviation or median, interquartile range for continuous data as appropriate and frequency, proportion for categorical data. We will check the representativeness of our sample by describing available characteristics for those patients who dropped out.

Faecal calprotectin concentration pre and post heart surgery will be described and compared using t tests. Changes in concentration levels will be derived and described and regression analysis will be used to explore the association between changes in concentration levels and the timing of the samples e.g. time of second sample relative to surgery time.

Logistic regression analysis will be used to assess the univariate relationship between calprotectin concentration (baseline, post-surgery and change) and NEC diagnosis; AUC, sensitivity and specificity will be presented, along with 95% CIs. Regression analysis will be performed to assess relationship between faecal calprotectin concentration and other factors such as weight, length of surgery (cardiopulmonary bypass) and antibiotic use.

4. Dissemination, Outputs and anticipated Impact

Outputs:

Our primary outputs will be:

- Ascertain whether there is a relationship between faecal calprotectin levels and necrotising enterocolitis and define calprotectin cut off levels that are clinically significant and sensitive to distinguish between NEC and No NEC.
- Assess the impact of the addition of a faecal calprotectin cut-off value to the current gold standard in diagnosing or 'ruling out' NEC in infants who were classified as suspected/ inconclusive NEC, to measure the patient burden of an erroneous diagnosis (nil by mouth/ antibiotic administration/ length of hospital stay)

Anticipated Impact

- Short term: Although infants and clinical staff will not experience a direct benefit or impact at this stage of the study, we feel by including this non-invasive intervention to the diagnostic discussion for NEC will continue to build an understanding of the biomarker and start to change the narrative around the clinical practice (stool

Pre-submission

collection) in infants with suspected NEC, which will support the validation follow on study

- Long term: Clinical impact and patient benefit potentially within 3 years. A follow-on study (outlined in more detail below in the future research section) to implement 'clinical calprotectin cut off levels for cardiac NEC' to assess the diagnostic accuracy of calprotectin when added to current diagnostic algorithm (Gold Standard - clinical presentation and abdominal X-ray) will mitigate an inconclusive suspected NEC diagnosis, therefore improving the timeliness and accuracy of making a NEC diagnosis.

Potential Patient Benefit:

- Avoid unnecessary withdrawal of an infant's milk feed, exacerbating parental stress in intensive care – optimise nutritional intake.
- Improved nutritional support has been associated with increased survival and a reduction in overall morbidity in infants with cardiac defects.
- Reduce unnecessary administration of antibiotics. Antibiotics cause antibiotic-related diarrhoea, which in turn affects expected growth and gut dysbiosis.
- Avoid inappropriate use of intravenous nutrition as a substitute for enteral feeds during gut rest. Avoiding intravenous nutrition reduces the risk of line sepsis

Wider health service impact

- Within 5 years: Update international clinical guidelines for the diagnosis of NEC with the addition of calprotectin cut off levels to improve the timeliness and accuracy of a NEC diagnosis (Association for European Paediatric and Congenital Cardiology – working group)
- Paediatric Critical Care Society (PCCS-UK) webinar
- European Teaching session Association for European Paediatric and Congenital Cardiology Basic courses and meetings | AEPC

Dissemination

National conference – Paediatric Critical Care Society: year 1 presentation to inform the UK about the project Paediatric; year 2 present preliminary findings (abstract) Home - Paediatric Critical Care Society (pccsociety.uk)

International conference - Association for European Paediatric and Congenital Cardiology – AEPC. Year 2 abstract/ oral presentation preliminary results About AEPC | AEPC

Publication: submit manuscript peer-reviewed journal

- NIHR End of Study Report
- Great Ormond Street Hospital, Institute of Child Health and University College London webpage via communication team
- GOSH Biomedical Research Centre dissemination to all the BRC in the United Kingdom
- Study Steering Group meeting to discuss future funding to trial calprotectin level in a clinical setting

Sharing results with participant families:

- End of study report - plain language summary to be published on NIHR website
- Within consent/ PIS will request families preferred contact method (family choice: email/ post) to receive summary of the findings acknowledging and respecting the contribution they have made. End of study information sheet/ newsletter
- Evaluate communication methods to establish how effective our dissemination strategy was, and whether it had the impact we wanted.

Future research - validation study: A more accurate test to confirm NEC will improve patient outcomes if the reductions in false positive or false negative results lead to more infants receiving the correct NEC diagnoses (diagnostic yield) and appropriate treatment (therapeutic yield). The degree to which appropriate treatment can improve patient outcomes depends on its efficacy (treatment efficacy). Although diagnostic yield generally increases with accuracy, it is also affected by a doctor's confidence in the diagnostic test.

Title: Assess the diagnostic accuracy of faecal calprotectin to 'rule out' NEC when added to the current diagnostic algorithm for NEC in infants with cardiac defects. Study Design: Test-treatment cluster randomised controlled trial. Include all 12 Paediatric Cardiac Units within the United Kingdom and Ireland.

Funding source: NIHR Health Technology Assessment (HTA) Programme

Methods and analysis 1:1 Randomisation. In this pragmatic cluster randomised controlled trial, we will randomise the 12 specialist cardiac units into intervention and control groups. The intervention group will add faecal calprotectin to its current diagnostic algorithm. The control group will test and treat according to current guidelines, which will not include faecal calprotectin. The medical team will include infants under three months old admitted to a specialist paediatric cardiac unit requiring cardiac surgery/ catheterisation.

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The primary outcome will be the number of infants diagnosed with suspected NEC as opposed to confirmed NEC. Clinical and economic burden of diagnostic error. Post study assessment of doctors' confidence to rule out or confirm NEC. A survey will be circulated to the medical professionals at each site who cared for infants with cardiac defects. The structured electronic online questionnaire will cover domains such as participant and practice demographic data, type of employment/practice, level of confidence in managing NEC (knowledge/skills), factors influencing diagnoses of NEC. A 5-point Likert scale will be used to assess the level of agreement between units who were randomised to add calprotectin to diagnostic algorithm.

5. Project / research timetable

Pre-study: HRA and IRAS ethical approval, establish Study Steering Group

Total study period: 24 months

Recruitment period: 18 months

Please see Gannt chart for detailed project timetable

6. Project Management

Co-lead Dr Graeme O'Connor and Prof Katherine Brown will take overall responsibility of the study ensuring key milestones are achieved including: HRA and REC Ethical approval for all four sites; recruitment target met; patient safety and appropriate recording of adverse events; representation at international committees to build interest and awareness of study. Directly supervise the study and take responsibility for study staff; chair SSG meetings. Subject safety - Protect subjects from harm.

Dr Graeme O'Connor will directly manage PPI Co-Applicant and the PPI Cardiac Parent Advisory Group (consisting of a PPI lead from each site – Alder Hay, Birmingham, GOSH and Southampton). Co-Chair with PPI Co-Applicant Lead Pre-SSG meetings, ensuring PPI team feel supported and prepared for SSG. Maintain records and site files.

Dr Graeme O'Connor will directly train and monitor all sites in relation to protocol adherence. Train qualified individuals to run the trial. Directly supervise the study and take responsibility for study staff. Bi-weekly virtual meetings with site leads to monitor recruitment and overall study adherence identify any problems.

The Study Steering Group (SSG) will meet a total of six times over 24 months. The SSG will consist of:

Great Ormond Street Hospital:
- Dr Graeme O'Connor (Joint Lead Applicant)
- Prof Katerine Brown (Joint Lead Applicant)
- Ms Deborah Ridout (Statistician)
- Mrs Hayley Gardner PPI Co-Applicant (Hayley.gardner88@outlook.com)

Alder Hey:
- Research Nurse lead Lauren Jones and Bernadette Khodaghalian (Consultant cardiologist)
- PPI lead TBC

Birmingham
- Research lead Carly Tooke and Helen Winmill
- PPI lead TBC

Southampton
- Prof Nigel Hall
- Research lead Nurse Jenny Pond
- PPI lead TBC

7. Ethics / Regulatory Approvals

GOSH research governance, HRA and ethical approval will be sought prior to study.

Potential ethical issues
- Infection risk during stool sample collection:

Mitigating action: the nurse caring for the infant should perform hand hygiene, wear gloves and apron. Comply with the infection prevention & control policies and procedures especially in relation to the Hand Hygiene Clinical Guideline, Standard and Isolation Precautions Policy and Uniform.

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8. Project / Research expertise

The joint lead applicant Professor Katherine Brown is an experienced Consultant in Paediatric Cardiac Intensive Care at Great Ormond Street Hospital and Deputy Head of Department of the Children's Institute of Cardiovascular Science at UCL. Professor Brown has experience leading national multicentre studies and has a far-reaching professional network and will supervise, mentor and support Dr O'Connor who is an early careers researcher. Dr O'Connor was the lead investigator in the pilot study that preceded this application and was supported in that role by Professor Brown.

PPI Co-Investigator: Mrs Gardner will be our expert lived experience parent who collaborated with Dr O'Connor during the pilot study and was instrumental in the design of our proposed study. Mrs Gardner's professional position working in a senior role in Human Resources has many transferable skills to support her PPI lead role including organisational, interpersonal, communication and kind. Mrs Gardner is also confident and assertive, willing to challenge researchers/ clinicians when necessary.

Co-Applicant Professor Nigel Hall (Southampton University Hospital) leads an active research program in children's surgical conditions with a specific focus on necrotising enterocolitis, other congenital gastrointestinal anomalies and the design and conduct of randomised trials on paediatric surgery. His team focusses on understanding the pathophysiology of necrotising enterocolitis and the. The development of novel therapies and other interventions aiming to improve short- and long-term outcome. Nationally he sits on the research committee of the British Association of Paediatric Surgeons. He is also Chair of the Network Office of the European Paediatric Surgeons Association

Co-Applicant Mrs Deborah Ridout (Institute of Child Health/ University College London) leads the Statistical Support Service (SSS), collaborating with colleagues throughout the Institute, UCL and with external academic and industry partners. Her main areas of research interest are paediatric neuromuscular disease and congenital heart disease. She is statistical editor of Neuromuscular Disorders. Mrs Ridout has been instrumental in developing the statistical component of the proposed study and will be a vital member of the SSG throughout the study period.

Collaborators at Partner sites

- **Alder Hey Children's Hospital:**

Dr Bernadette Khodaghalian (consultant cardiologist at Alder Hay) research experience in pathogenesis of a disease in a preterm infant with coarctation of the aorta. Also completed a service evaluation to monitor cardiac NEC using calprotectin as a marker of gut disease: Bernadette.Khodaghalian@alderhey.nhs.uk

Head of Cardiology: Dr Caroline Jones: Caroline.Jones2@alderhey.nhs.uk

Research Nurse Lead: Mrs Laura Rad: Laura.Rad@alderhey.nhs.uk

- **Birmingham Childrens Hospital:**

Intensive care research Lead: Mrs Helen Winmill: Helen <h.winmill@nhs.net>

Research nurse: Mrs Samantha Owen: samantha.owen10@nhs.net

- **Southampton Hospital:**

Southampton: Consultant surgeon Prof Nigel Hall: n.j.hall@soton.ac.uk

Research lead nurse. Mrs Jenny Pond: j.pond@soton.ac.uk

9. Success criteria and barriers to proposed work

Success criteria 1: deferred consent proves effective with no/minimal parent withdrawal from the study. This will be emphasised during the training with a brief invitation outlining the study followed by PIS and Co-I explanation – highlight non-invasive intervention with significant potential benefit for future work.

Success criteria 2: Calprotectin levels remain hidden from clinical staff to ensure medical treatment is not altered.

Success criteria 3: meet expected recruitment rate 4-5 infants per month per site (16-20 infants per month)

Possible barrier for further study:

Challenge 1. Calprotectin analysis and availability of results at ward level: real time turnaround needs to be timely to have a meaningful impact (24-48hrs). All four sites have a turnaround time of 24hrs. Calprotectin concentration

Pre-submission

measurements is standard practice in other areas of medicine such as gastroenterology.

Challenge 2. Availability of stool when NEC is suspected – having to wait for spontaneous bowel movement or stools are too loose/ watery to collect. During our pilot study this accounted for around 3% of cohort (Mitigating action – accounted for in attrition rate).

Challenge 3. At each site calprotectin levels will be hidden from clinicians to ensure they do not change their usual treatment in infants with suspected NEC. (Mitigating actions: liaise with laboratory's at each centre to ensure calprotectin levels can only be retrieved by the research nurse)

	Pre-study	Year 1 (2025/26)												Year 2 (2026/27)												Post-study
ect agement		M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	
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7. Patient & Public Involvement

Please describe how patients and the public have been involved in developing this proposal.

I led two focus groups which consisted of parents whose infants were on the cardiac intensive care unit at Great Ormond Street (total of 9 parents). Open ended questions were asked around: how important is nutrition to you in respect to all other cardiac problems; how often has your baby's feed been stopped; are you aware of the gut complications associated with cardiac problems.

Parent focus group outcomes:

- All parents rated nutrition as being as important as medical intervention. Furthermore, parents clearly linked nutrition and growth with overall wellbeing in relation to cardiac health and convalescents, and all had experience of feeds being withdrawn for one reason or another. However, the vast majority of parents were not aware of the implications their child's cardiac defect had on their gut.

- Deferred consent and appropriate time for researchers to approach families to discus study. Interestingly, parents felt it was acceptable to be approached soon after their baby was admitted (24-48hrs) as most parents received an antenatal diagnosis of their baby's cardiac defects and therefore although a stressful time post-delivery felt

- Tube feeding – parents thought it was important to know how long it would be until they could breast/ bottle feed their child. Length of time to botte/ breast fed post-surgery has been included to study.

- Parents used the term 'starving my baby', when feed are withdrawn for medical investigations/ gut rest, which provided further insight into how emotive feeding is for parents.

- Medical terminology - gut disease was perceived to be an acceptable term to describe necrotising enterocolitis

Please describe the ways in which patients and the public will be actively involved in the proposed research, including any training and support provided.

PPI Team

PPI Co-Applicant Mrs Hayley Gardner: expert parent with lived experience of child with cardiac defect.

PPI Cardiac Advisory Group: at least one parent representative from all four centres. Lived experience of having an infant in intensive care or an young person/ adult with a heart defect.

Roles:

- To attend all Study Steering Group (SSG) meetings: This will include nine meetings over the study period. The PPI Co-Applicant and PPI Cardiac Advisory Group will also attend a Pre SSG meeting to ensure the PPI team is prepared for SSG. The pre-SSG meetings will be chaired by the PPI Co-Applicant.

- SSG meeting preparation: completing reviews of documents (Lay summary/ patient information sheets/ end of study summary report for participating families).

- Unforeseen study problems may need the PPI Team to participate in a focus group to find a solution to problem.

For involvement in a task or activity where preparation is required and which equates to approximately half a day's activity. For example, participating in a meeting to interview candidates who have applied to join a committee, participating in a focus group, or delivering training.

Training:

Training course: Homepage - Learning for Involvement

NIHR Guidance on co-producing a research project (learningforinvolvement.org.uk)

NIHR Co-Production in Action (One, two, three and four) - Learning for Involvement

Pre-submission

In-house staff development and training events related to the study topic.

Support:

- Financial: Mrs Gardiner (PPI Co-Applicant) will be registered with an an Great Ormond Street Honorary contract and will be registered as a Supplier and can therefore invoice the CI for an electronic payment for cover time. Members of the PP Cardiac Advisory Team will receive a voucher (£60 per SSG Meeting) to cover their time for pre-reading and SSG meetings

- Assessing and reporting progress of the PPI Teams input will be provided using the Elspeth Mathie facilitating feedback model.

- Support outside of the research team will include: Deirdre Leyden PPI Manager at GOSH and INVOLVE.

We will ensure the PPI Team is representative of a diverse population (ethnicity and social economic class)

In rare cases where proposals do NOT involve patients and the public, clear justification must be provided.
NA

8. Detailed Budget

Note: The SoECAT Form is included as a separate document.

Staff Costs

Budget Item 1

Staff Posts and Salaries

Staff name	Graeme O'Connor

Staff type	Lead & Co-applicant

Organisation name	Great Ormond Street Hospital for Children NHS Foundation Trust

Organisation type	NHS (UK)

Role	Research lead - study management of all sites

Grade	8a

Salary (£)
60,504

On-costs (£)
33,023

Current annual salary (£)	93,527.00

Pre-submission

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
100

Base annual salary with rate	£93,527.00

	Year 1	Year 2	
The year 1 value should always be 0%.	0	3	

	Year 1	Year 2	
FTE	30	30	

	Year 1	Year 2	
Months	12	12	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£28,058.00	£28,900.00	£56,958.00

Budget Item 2

Staff Posts and Salaries

Staff name	Katherine Brown

Staff type	Lead & Co-applicant

Organisation name	Great Ormond Street Hospital for Children NHS Foundation Trust

Organisation type	NHS (UK)

Role	consultant cardiologist study management support

Grade	Medical Consultant YM

Salary (£)
131,964

On-costs (£)

Pre-submission

51,936

Current annual salary (£)	183,900.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
100

Base annual salary with rate	£183,900.00

	Year 1	Year 2	
The year 1 value should always be 0%.	0	3	

	Year 1	Year 2	
FTE	10	10	

	Year 1	Year 2	
Months	12	12	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£18,390.00	£18,942.00	£37,332.00

Budget Item 3

Staff Posts and Salaries

Staff name	Nigel Hall

Staff type	Lead & Co-applicant

Organisation name	University Hospital Southampton NHS Foundation Trust

Organisation type	NHS (UK)

Role	Consultant Paediatric Surgeon

Grade	MC72

Pre-submission

Salary (£)
118,884

On-costs (£)
32,389

Current annual salary (£)	151,273.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
100

Base annual salary with rate	£151,273.00

	Year 1	Year 2	
The year 1 value should always be 0%.	0	1	

	Year 1	Year 2	
FTE	5	5	

	Year 1	Year 2	
Months	6	12	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£3,782.00	£7,639.00	£11,421.00

Budget Item 4

Staff Posts and Salaries

Staff name	Research Nurse Birmingham Childrens

Staff type	Research Staff

Organisation name	Birmingham Children's Hospital

Organisation type	NHS (UK)

Pre-submission

Role	Research lead for Birmingham children

Grade	Afc Band 7

Salary (£)
48,526

On-costs (£)
20,461

Current annual salary (£)	68,987.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
100

Base annual salary with rate	£68,987.00

	Year 1	**Year 2**	
The year 1 value should always be 0%.	0	2	

	Year 1	**Year 2**	
FTE	15	15	

	Year 1	**Year 2**	
Months	6	12	

	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Total rate calculation	£5,174.00	£10,555.00	£15,729.00

Budget Item 5

Staff Posts and Salaries

Staff name	Research Nurse Alder Hay Childrens

Staff type	Research Staff

Pre-submission

Organisation name	Alder Hey Children's Hospital

Organisation type	NHS (UK)

Role	Research lead for study

Grade	Afc Band 7

Salary (£)
48,526

On-costs (£)
20,000

Current annual salary (£)	68,526.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
100

Base annual salary with rate	£68,526.00

	Year 1	**Year 2**	
The year 1 value should always be 0%.	0	2	

	Year 1	**Year 2**	
FTE	15	15	

	Year 1	**Year 2**	
Months	6	12	

	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Total rate calculation	£5,139.00	£10,484.00	£15,623.00

Budget Item 6

Staff Posts and Salaries

Pre-submission

Staff name	Research Nurse Southampton

Staff type	Research Staff

Organisation name	University Hospital Southampton NHS Foundation Trust

Organisation type	NHS (UK)

Role	Research lead for study

Grade	Afc Band 7

Salary (£)
48,500

On-costs (£)
20,000

Current annual salary (£)	68,500.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
100

Base annual salary with rate	£68,500.00

	Year 1	**Year 2**	
The year 1 value should always be 0%.	0	2	

	Year 1	**Year 2**	
FTE	15	15	

	Year 1	**Year 2**	
Months	6	12	

	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Total rate calculation	£5,138.00	£10,481.00	£15,619.00

Pre-submission

Budget Item 7

Staff Posts and Salaries

Staff name	Deborah Ridout

Staff type	Lead & Co-applicant

Organisation name	NIHR Great Ormond Street Biomedical Research Centre

Organisation type	HEI (UK)

Role	Biostatistician

Grade	8b

Salary (£)
60,000

On-costs (£)
10,000

Current annual salary (£)	70,000.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
80

Base annual salary with rate	£56,000.00

	Year 1	**Year 2**	
The year 1 value should always be 0%.	0	2	

	Year 1	**Year 2**	
FTE	10	10	

	Year 1	**Year 2**	
Months	6	12	

Pre-submission

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£2,800.00	£5,712.00	£8,512.00

Travel, Subsistence & Conference Costs

Budget Item 1

Item description
Train travel to three sites for training and monitorin

Organisation Name	Great Ormond Street Hospital for Children NHS Foundation Trust

Type of Cost	Travel

Organisation Type	NHS (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£600.00	£400.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£600.00	£400.00	£1,000.00

Budget Item 2

Item description
Conference fees The 59th Annual Meeting of the Association for European Paediatric and Congenital Cardiology - AEPC 2026

Organisation Name	Great Ormond Street Hospital for Children NHS Foundation Trust

Type of Cost	Conference

Organisation Type	NHS (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£0.00	£600.00	

Pre-submission

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£0.00	£600.00	£600.00

Equipment Costs

Consumables

Budget Item 1

Item description	Calprotectin Analysis

Organisation Name	Great Ormond Street Hospital for Children NHS Foundation Trust

Organisation Type	NHS (UK)

Rate %	100

	Year 1	Year 2	
Costs	£18,000.00	£18,000.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£18,000.00	£18,000.00	£36,000.00

Patient & Public Involvement, Engagement and Participation Costs

Budget Item 1

Item description	PPI Co-Applicant and PPI lead for GOSH

Organisation Name	Great Ormond Street Hospital for Children NHS Foundation Trust

Organisation Type	NHS (UK)

Rate %	100

	Year 1	Year 2	
Costs	£1,500.00	£1,500.00	

Pre-submission

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£1,500.00	£1,500.00	£3,000.00

Budget Item 2

Item description	PPI leads at 3 partner sites

Organisation Name	Great Ormond Street Hospital for Children NHS Foundation Trust

Organisation Type	NHS (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£550.00	£550.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£550.00	£550.00	£1,100.00

Other Direct Costs

Budget Item 1

Item description	ISRCTN registration fee

Organisation Name	Great Ormond Street Hospital for Children NHS Foundation Trust

Organisation Type	NHS (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£300.00	£0.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£300.00	£0.00	£300.00

Pre-submission

Budget Item 2

Item description	Multisite REDCAP

Organisation Name	Great Ormond Street Hospital for Children NHS Foundation Trust

Organisation Type	NHS (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£2,000.00		

	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Total rate calculation	£2,000.00	£0.00	£2,000.00

Dissemination Costs

Budget Item 1

Item description
Travel/ accommodation The 59th Annual Meeting of the Association for European Paediatric and Congenital Cardiology - AEPC 2026

Organisation Name	Great Ormond Street Hospital for Children NHS Foundation Trust

Organisation Type	NHS (UK)

Rate %	100

	Year 1	**Year 2**	
Costs		£1,500.00	

	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Total rate calculation	£0.00	£1,500.00	£1,500.00

Indirect Costs

Pre-submission

Budget Item 1

Item description	Interpreter Service https://en-gb.thebigword.com/

Organisation Name	National Health Service

Type of Cost	Other Indirect Costs

Organisation Type	Commercial (UK)

Rate % Please note for HEIs in the UK, up to 80% of FEC will be paid.
100

	Year 1	**Year 2**	
Costs	£300.00	£300.00	

	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Total rate calculation	£300.00	£300.00	£600.00

Justification of costs
1. Staff costs: G O'Connor joint lead applicant - 24months 30% FTE Overall responsibility for day to day management of the trial at all four sites. Support PPI leads for each sites Data collection for Great Ormond Street ~ 75 infants Lead Ethical approval Prof Katherine Brown 10% for Joint Lead Overall support for the delivery of study. Supervision throughout study period as per Gantt Chart Co-Applicants: 10% statistician - 18 months: support to meet objectives to ascertain if calprotectin is a marker of gut disease. Area under curve analysis and linear and logistic regression. Support to develop future study 5% Independent adjudicator to review 40-50 patient records to confirm gut disease 15% Band 7 Research lead nurse at each site - 18 months: Birmingham, Alder Hey and Southampton: 75 infants per site data collection. 2. Travel, Subsistence & Conference Costs Travel to three sites for training and monitoring three times over the study period as per Gantt chart International conference: registration fees/ flights and accommodation - dissemination of results 2026 3. Consumables Calprotectin: cost to analyse stool samples to measure calprotectin concentrations: 600 samples at £60 each Stool collection pots x 600 4. PPIEP Costs PPI Co-applicants and PPI lead for Great Ormond Street: Hayley Gardner £1245 PPI Co-applicant: 6 SSG meetings over 24 months: 6 x £150 (£900) This includes preparation and attendance for one day meetings. Direct time: 3hr per meeting. Indirect time: 2-3hrs per meeting additional reading Private Childcare – (£50 x 6 meetings = £300) £5 per meeting to cover the cost of telephone calls, paper, printing ink

Pre-submission

and paper, internet connection and other home sundries (£45)

Other PPI Costs: Cardiac Parent Advisory Group (PPI Leads at each site Birmingham, Alder Hey and Southampton): 3 site leads x £60 x 6 vouchers per meeting including pre-reading/ preparation (£1080)

5. Other Direct Costs
REDCAP research data collection portal support service for a multisite study. Support include study set up, IT support throughout project and formatted for statistical analysis.

6. Dissemination Costs
The 59th Annual Meeting of the Association for European Paediatric and Congenital Cardiology - AEPC 2026. Travel to Europe and 3 nights accommodation.

Totals			
	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Staff Costs	£68,481.00	£92,713.00	£161,194.00
Travel, Subsistence & Conference Costs	£600.00	£1,000.00	£1,600.00
Equipment Costs	£0.00	£0.00	£0.00
Consumables	£18,000.00	£18,000.00	£36,000.00
PPIEP Costs	£2,050.00	£2,050.00	£4,100.00
Other Direct Costs	£2,300.00	£0.00	£2,300.00
Dissemination Costs	£0.00	£1,500.00	£1,500.00
Indirect Costs	£300.00	£300.00	£600.00
Total	**£91,731.00**	**£115,563.00**	**£207,294.00**

9. Management & Governance

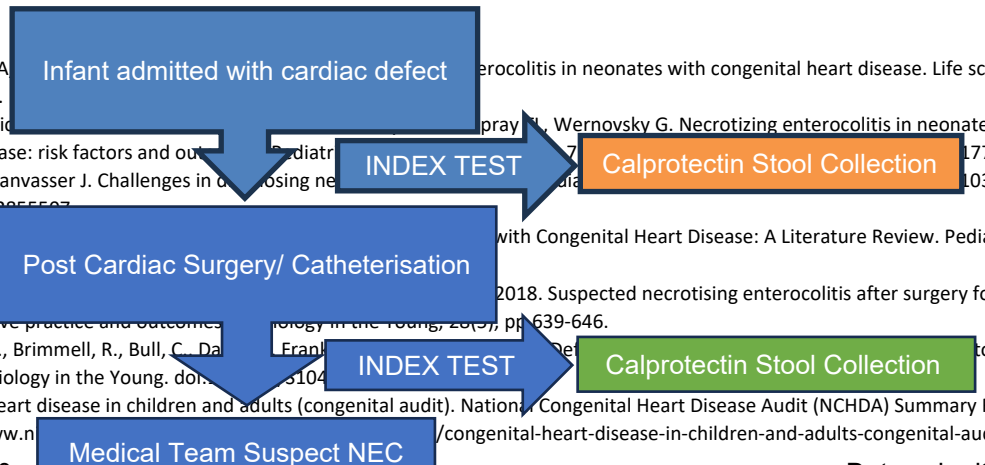
Is Clinical Trials Authorisation required?	No

Does your project require ethics approval?	Yes

If yes, has ethics approval already been obtained?	No

10. Uploads

Study Flow Chart



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Pre-submission

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11. Administrative contact details

Administrative contact name	Laura Lucas

Administrative contact job title	Research Grants Facilitator

Pre-submission

Administrative contact telephone number	020 7905 2700

Administrative contact email address	Research.Registration@gosh.nhs.uk

12. Research & Development office contact

R&D contact name	Dr Vanshree Patel

R&D contact job title	Head of Research and Developement

R&D contact telephone number	07395 852101

R&D contact email address	research.governance@gosh.nhs.uk

13. Acknowledgement, review and submit

COI declaration Please declare any conflicts or potential conflicts of interest that you or your co-applicants may have in undertaking this research, including any relevant personal, non-personal and commercial interest that could be perceived as a conflict of interest.
No conflict of interest to declare

Lead Applicant - Agreement to terms and conditions

In **ticking** this, you as **Lead Applicant** confirm that the information given on this form is correct and that you will be actively engaged in this research and responsible for its overall management. In addition, you will accept responsibility for ensuring that the host institution and interested parties are kept informed. **Ticking this box constitutes an electronic signature of the Lead Applicant with regard to this application**
☒Confirmed

Appropriate and relevant involvement of patients/service users, carers and the public
☒Confirmed

A good quality plain English summary	☒Confirmed

A clear description of team member roles and contribution
☒Confirmed

A clear scientific abstract
☒Confirmed

A clear description of the changes from first stage
☒Confirmed

Pre-submission

A flow diagram illustrating the study design / flow of participants (document upload), if appropriate
☒Confirmed

A full and accurate detailed budget breakdown
☒Confirmed

A clear justification of costs / value for money
☒Confirmed

References (document upload)
☒Confirmed

A clear Detailed Research Plan outlining the study design, methods, dissemination etc.
☒Confirmed

A CTU letter of support if required (document upload)
☒Confirmed

Completed Schedule of Events Cost Attribution Tool (SoECAT)
☒Confirmed

14. CV - Lead Applicant(s)

Full Name	Dr GRAEME O'CONNOR	**Position**	Paediatric Dietitian
Department	Intensive Care	**Institution**	Great Ormond Street Hospital for Children NHS Foundation Trust
ORCID iD	0000-0001-8625-9264	**Telephone No.**	+44 7958543828
Address Line 1	Level 2 Nurses Building	**Country**	United Kingdom
Address Line 2	Great Ormond Street	**Postcode**	WC1N 3JH
Address Line 3		**Web Page.**	

Qualification	Subject	Institution	From	To
PhD	Refeeding and malnutrition	UCL	September 2011	March 2014
Post Graduate Dipolma	First Contact Practitioner	Sheffield Hallam University	August 2006	June 2008
MSc	Dietetics	Glasgow Caledonian University	September 2000	September 2002
BSc	Physiology	University of Greenwich	September 1996	September 2000

Selected recent relevant publication(s)

Publications
(2020) 'Faecal calprotectin concentrations in neonates with CHD: pilot study', *Cardiology in the young*, (5):624-628(30).

Pre-submission

Selected recent relevant publication(s)

Publications
(2022) 'Research Priorities for Pediatric Intensive Care Nutrition Within the United Kingdom: A National Institute of Health Research James Lind Alliance Priority Setting Partnership', *Critical care explorations*, 4(3):e0649.(4).
(2024) 'An open-label pilot single-subject study to monitor the impact of a Food-Based enteral formula on faecal short-chain fatty acid concentrations in children admitted to intensive care with sepsis',
(2024) 'The impact of low-energy, partially hydrolysed enteral formula on gastrointestinal symptoms and weight in children with neurological impairment: a multicentre retrospective study',
(2021) 'Monitor gastrointestinal tolerance in children who have switched to an "enteral formula with food-derived ingredients": A national, multicenter retrospective chart review (RICIMIX study)', *Nutrition in clinical practice : official publication of the American Society for Parenteral and Enteral Nutrition*, 37(4).
(2014) 'Refeeding low weight hospitalised adolescents with anorexia nervosa: national multi centre Randomised Controlled Trial.',

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant
01/01/2024	12	80,000.00	Nestle Health Science	Prospective case control study to monitor nutritional status and intestinal microbiome of children undergoing acute myeloid leukaemia treatment: case control study	Principal Investigator
01/06/2023	12	6,000.00	University of Central Lancashire ARC Grant	Creative participatory method to explore children's experiences in intensive care: post intensive care syndrome	Joint Lead Principal Investigator

Joint Lead Applicant

Full Name Professor Katherine Brown

Institution NIHR Great Ormond Street Biomedical Research Centre

Email Address katherine.brown@gosh.nhs.uk

Web Page.

Education

From	To	Qualification	Subject	Country	Organisation	Department
06/2012	06/2015	MD	Development and initial applications of a risk model to adjust for severity of case mix in paediatric cardiac surgery using the national audit database	United Kingdom	Cambridge University	Clinical Medical School
09/2006	06/2008	MPH	Public Health Health Services Research	United Kingdom	London School of Hygiene & Tropical Medicine	Public Health and Policy
05/1993	06/1995	MRCP	MRCP	United Kingdom (England)	Royal College of Physicians England	Royal College of Physicians

Pre-submission

Education

From	To	Qualification	Subject	Country	Organisation	Department
						London
09/1989	06/1992	BA	Medical Sciences	United Kingdom	Cambridge University	Caius College
09/1989	06/1992	MBBChir	Medicine and Surgery	United Kingdom	Cambridge University	Caius College

Publications

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant

15. CV - Co-applicants

Co Applicant - 1

Full Name Professor Nigel Hall
Email Address n.j.hall@soton.ac.uk
Organisation University of Southampton
Web Page.

Education

From	To	Qualification	Subject	Country	Organisation	Department
06/2010	09/2010	FRCS (Paed.Surg)	Paediatric Surgery		Royal College of Surgeons of England	N/A
10/2002	10/2005	PhD	Paediatric Surgery		University of London	Institute of Child Health
06/1999	07/2000	MRCPCH	Paediatrics		Royal College of Paediatrics and Child Health	Paediatrics
09/1992	12/1997	MBBChir	Clinical Medicine		University of Cambridge	Medical School

Publications

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant

Co Applicant - 2

Full Name Ms Deborah Ridout
Email Address d.ridout@ucl.ac.uk
Organisation UCL Great Ormond Street Institute of Child Health
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Education

From	To	Qualification	Subject	Country	Organisation	Department
09/1989	09/1990	MSc	Statistics	United Kingdom	University of Kent	Statistics
09/1985	06/1989	BSc	Mathematics, Statistics & Computing	United Kingdom	University of Greenwich	Mathematics & Statistics

Pre-submission

Publications

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant

Pre-submission